MSG No. 142.570 I_c4_1/acd [Type IV, tetragonal]

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det	TR
1	$\{1\|0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
2	$\{4^+_{001}\|\frac{1}{4}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{4} \end{bmatrix}$	1	1
3	$\{4^-_{001}\|\frac{3}{4}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{4} \end{bmatrix}$	1	1
4	$\{2_{100}\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	1
5	$\{2_{010}\|\frac{1}{2}00\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	1
6	$\{2_{001}\|0\frac{1}{2}0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
7	$\{2_{110}\|\frac{3}{4}\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{3}{4} \\ 1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	1	1
8	$\{2_{1-10}\|\frac{1}{4}\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	1	1
9	$\{-1\|0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	1
10	$\{-4^+_{001}\|\frac{3}{4}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{3}{4} \end{bmatrix}$	−1	1
11	$\{-4^-_{001}\|\frac{1}{4}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{3}{4} \end{bmatrix}$	−1	1
12	$\{\mathrm{m}_{100}\|00\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	−1	1
13	$\{\mathrm{m}_{010}\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	−1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
14	$\{\mathrm{m}_{001}\|0\frac{1}{2}0\}$	$\begin{bmatrix}1&0&0&0\\0&1&0&\frac{1}{2}\\0&0&-1&0\end{bmatrix}$	-1	1
15	$\{\mathrm{m}_{110}\|\frac{3}{4}\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix}0&-1&0&\frac{3}{4}\\-1&0&0&\frac{1}{4}\\0&0&1&\frac{1}{4}\end{bmatrix}$	-1	1
16	$\{\mathrm{m}_{1-10}\|\frac{1}{4}\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix}0&1&0&\frac{1}{4}\\1&0&0&\frac{1}{4}\\0&0&1&\frac{1}{4}\end{bmatrix}$	-1	1
17	$\{1\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}1&0&0&\frac{1}{2}\\0&1&0&\frac{1}{2}\\0&0&1&\frac{1}{2}\end{bmatrix}$	1	1
18	$\{4^{+}_{001}\|\frac{3}{4}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix}0&-1&0&\frac{3}{4}\\1&0&0&\frac{1}{4}\\0&0&1&\frac{3}{4}\end{bmatrix}$	1	1
19	$\{4^{-}_{001}\|\frac{1}{4}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix}0&1&0&\frac{1}{4}\\-1&0&0&\frac{1}{4}\\0&0&1&\frac{3}{4}\end{bmatrix}$	1	1
20	$\{2_{100}\|00\frac{1}{2}\}$	$\begin{bmatrix}1&0&0&0\\0&-1&0&0\\0&0&-1&\frac{1}{2}\end{bmatrix}$	1	1
21	$\{2_{010}\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}-1&0&0&0\\0&1&0&\frac{1}{2}\\0&0&-1&\frac{1}{2}\end{bmatrix}$	1	1
22	$\{2_{001}\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&-1&0&0\\0&0&1&\frac{1}{2}\end{bmatrix}$	1	1
23	$\{2_{110}\|\frac{1}{4}\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix}0&1&0&\frac{1}{4}\\1&0&0&\frac{3}{4}\\0&0&-1&\frac{3}{4}\end{bmatrix}$	1	1
24	$\{2_{1-10}\|\frac{3}{4}\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix}0&-1&0&\frac{3}{4}\\-1&0&0&\frac{3}{4}\\0&0&-1&\frac{3}{4}\end{bmatrix}$	1	1
25	$\{-1\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&-1&0&\frac{1}{2}\\0&0&-1&\frac{1}{2}\end{bmatrix}$	-1	1
26	$\{-4^{+}_{001}\|\frac{1}{4}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix}0&1&0&\frac{1}{4}\\-1&0&0&\frac{3}{4}\\0&0&-1&\frac{1}{4}\end{bmatrix}$	-1	1
27	$\{-4^{-}_{001}\|\frac{3}{4}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix}0&-1&0&\frac{3}{4}\\1&0&0&\frac{3}{4}\\0&0&-1&\frac{1}{4}\end{bmatrix}$	-1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
28	$\{\mathrm{m}_{100}\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}-1 & 0 & 0 & \frac{1}{2}\\ 0 & 1 & 0 & \frac{1}{2}\\ 0 & 0 & 1 & 0\end{bmatrix}$	-1	1
29	$\{\mathrm{m}_{010}\vert\frac{1}{2}00\}$	$\begin{bmatrix}1 & 0 & 0 & \frac{1}{2}\\ 0 & -1 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}$	-1	1
30	$\{\mathrm{m}_{001}\vert\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix}1 & 0 & 0 & \frac{1}{2}\\ 0 & 1 & 0 & 0\\ 0 & 0 & -1 & \frac{1}{2}\end{bmatrix}$	-1	1
31	$\{\mathrm{m}_{110}\vert\frac{1}{4}\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix}0 & -1 & 0 & \frac{1}{4}\\ -1 & 0 & 0 & \frac{3}{4}\\ 0 & 0 & 1 & \frac{3}{4}\end{bmatrix}$	-1	1
32	$\{\mathrm{m}_{1-10}\vert\frac{3}{4}\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix}0 & 1 & 0 & \frac{3}{4}\\ 1 & 0 & 0 & \frac{3}{4}\\ 0 & 0 & 1 & \frac{3}{4}\end{bmatrix}$	-1	1
33	$\{1'\vert00\frac{1}{2}\}$	$\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & \frac{1}{2}\end{bmatrix}$	1	-1
34	$\{4^{+}_{001}{}'\vert\frac{1}{4}\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix}0 & -1 & 0 & \frac{1}{4}\\ 1 & 0 & 0 & \frac{3}{4}\\ 0 & 0 & 1 & \frac{3}{4}\end{bmatrix}$	1	-1
35	$\{4^{-}_{001}{}'\vert\frac{3}{4}\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix}0 & 1 & 0 & \frac{3}{4}\\ -1 & 0 & 0 & \frac{3}{4}\\ 0 & 0 & 1 & \frac{3}{4}\end{bmatrix}$	1	-1
36	$\{2_{100}{}'\vert\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}1 & 0 & 0 & \frac{1}{2}\\ 0 & -1 & 0 & \frac{1}{2}\\ 0 & 0 & -1 & \frac{1}{2}\end{bmatrix}$	1	-1
37	$\{2_{010}{}'\vert\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix}-1 & 0 & 0 & \frac{1}{2}\\ 0 & 1 & 0 & 0\\ 0 & 0 & -1 & \frac{1}{2}\end{bmatrix}$	1	-1
38	$\{2_{001}{}'\vert0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}-1 & 0 & 0 & 0\\ 0 & -1 & 0 & \frac{1}{2}\\ 0 & 0 & 1 & \frac{1}{2}\end{bmatrix}$	1	-1
39	$\{2_{110}{}'\vert\frac{3}{4}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix}0 & 1 & 0 & \frac{3}{4}\\ 1 & 0 & 0 & \frac{1}{4}\\ 0 & 0 & -1 & \frac{3}{4}\end{bmatrix}$	1	-1
40	$\{2_{1-10}{}'\vert\frac{1}{4}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix}0 & -1 & 0 & \frac{1}{4}\\ -1 & 0 & 0 & \frac{1}{4}\\ 0 & 0 & -1 & \frac{3}{4}\end{bmatrix}$	1	-1
41	$\{-1'\vert00\frac{1}{2}\}$	$\begin{bmatrix}-1 & 0 & 0 & 0\\ 0 & -1 & 0 & 0\\ 0 & 0 & -1 & \frac{1}{2}\end{bmatrix}$	-1	-1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
42	$\{-4^{+}_{001}{}'\vert\frac{3}{4}\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	-1	-1
43	$\{-4^{-}_{001}{}'\vert\frac{1}{4}\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	-1	-1
44	$\{\mathrm{m}_{100}{}'\vert 0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	-1	-1
45	$\{\mathrm{m}_{010}{}'\vert 0\frac{1}{2}0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	-1	-1
46	$\{\mathrm{m}_{001}{}'\vert 0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	-1	-1
47	$\{\mathrm{m}_{110}{}'\vert\frac{3}{4}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}$	-1	-1
48	$\{\mathrm{m}_{1-10}{}'\vert\frac{1}{4}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}$	-1	-1
49	$\{1'\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	-1
50	$\{4^{+}_{001}{}'\vert\frac{3}{4}\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{3}{4} \\ 1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{4} \end{bmatrix}$	1	-1
51	$\{4^{-}_{001}{}'\vert\frac{1}{4}\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{4} \end{bmatrix}$	1	-1
52	$\{2_{100}{}'\vert 0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	-1
53	$\{2_{010}{}'\vert 0\frac{1}{2}0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	-1
54	$\{2_{001}{}'\vert\frac{1}{2}00\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	-1
55	$\{2_{110}{}'\vert\frac{1}{4}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	1	-1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
56	$\{2_{1-10}{}'\vert\frac{3}{4}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	1	−1
57	$\{-1'\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	−1
58	$\{-4^{+}_{001}{}'\vert\frac{1}{4}\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{3}{4} \end{bmatrix}$	−1	−1
59	$\{-4^{-}_{001}{}'\vert\frac{3}{4}\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{3}{4} \\ 1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{3}{4} \end{bmatrix}$	−1	−1
60	$\{\mathrm{m}_{100}{}'\vert\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	−1	−1
61	$\{\mathrm{m}_{010}{}'\vert\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	−1	−1
62	$\{\mathrm{m}_{001}{}'\vert\frac{1}{2}00\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	−1
63	$\{\mathrm{m}_{110}{}'\vert\frac{1}{4}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{4} \end{bmatrix}$	−1	−1
64	$\{\mathrm{m}_{1-10}{}'\vert\frac{3}{4}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{3}{4} \\ 1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{4} \end{bmatrix}$	−1	−1